

```

data raw_dm;
  length USUBJID $10 ARM $20 SEX $1 RACE $25;
  input USUBJID $ ARM $ AGE SEX $ RACE $;
  datalines;
SUBJ-001 Drug_A_100mg 45 F White
SUBJ-002 Placebo 52 M Black_or_African_American
SUBJ-003 Drug_A_100mg 68 F Asian
SUBJ-004 Placebo 34 M White
SUBJ-005 Drug_A_100mg 29 F White
SUBJ-006 Placebo 61 F White
SUBJ-007 Drug_A_100mg 55 M Black_or_African_American
SUBJ-008 Placebo 72 M Asian
SUBJ-009 Drug_A_100mg 41 F White
SUBJ-010 Placebo 50 F Asian
SUBJ-011 Drug_A_100mg 38 M White
SUBJ-012 Placebo 64 M Black_or_African_American
SUBJ-013 Drug_A_100mg 59 F White
SUBJ-014 Placebo 47 F Asian
SUBJ-015 Drug_A_100mg 53 M White
SUBJ-016 Placebo 31 M White
SUBJ-017 Drug_A_100mg 66 F White
SUBJ-018 Placebo 49 M Asian
SUBJ-019 Drug_A_100mg 58 M Black_or_African_American
SUBJ-020 Placebo 43 F White
;
run;

```

```

data raw_ae;
  length USUBJID $10 AETERM $40 AESEV $10 AESTDTC $10 AEENDTC $10;
  input USUBJID $ AETERM $ AESEV $ AESTDTC $ AEENDTC $;
  datalines;
SUBJ-001 Headache Mild 2026-01-10 2026-01-12
SUBJ-001 Nausea Severe 2026-01-15 2026-01-16
SUBJ-003 Dizziness Moderate 2026-02-01 2026-02-03
SUBJ-005 Fatigue Mild 2026-01-20 2026-01-22
SUBJ-007 Rash Moderate 2026-03-05 2026-03-10
SUBJ-008 Headache Severe 2026-02-14 2026-02-18
SUBJ-009 Nausea Mild 2026-01-11 2026-01-12
SUBJ-011 Dizziness Moderate 2026-03-01 2026-03-02
SUBJ-013 Fatigue Severe 2026-02-20 2026-02-25
SUBJ-015 Headache Mild 2026-01-05 2026-01-07
SUBJ-017 Rash Severe 2026-03-12 2026-03-15
SUBJ-019 Dizziness Moderate 2026-02-28 2026-03-01
;
run;

```

```

data adsl;
  set raw_dm;
  if AGE < 50 then AGEGR1 = "<50 Years";
  else if AGE >= 50 and AGE <= 65 then AGEGR1 = "50-65 Years";

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    else if AGE > 65 then AGEGR1 = ">65 Years";
    SAFFL = "Y";
run;

proc sort data=adsl; by USUBJID; run;
proc sort data=raw_ae; by USUBJID; run;

data adae;
  merge adsl(in=a) raw_ae(in=b);
  by USUBJID;
  if a and b;
  if not missing(AETERM) then TRTEMFL = 'Y';
  else TRTEMFL = 'N';
run;

proc univariate data=adsl normal;
  var AGE;
  id USUBJID;
  title "Data Quality Check: Age Distribution Analysis";
run;

proc freq data=adsl;
  tables ARM * (SEX RACE AGEGR1) / list missing;
  title "Data Quality Check: Categorical Demographics Cross-Tabulation";
run;

proc report data=adsl headline headskip split='*';
  column ARM AGEGR1 SEX AGE;
  define ARM / group "Treatment Arm" order=internal;
  define AGEGR1 / group "Age Group";
  define SEX / group "Gender";
  define AGE / analysis mean format=5.1 "Mean Age (Years)";
  title "Table 14.1.1: Demographic and Baseline Characteristics";
run;

proc sql;
  create table ae_summary as
  select ARM,
         AETERM,
         count(distinct USUBJID) as Subject_Count,
         (calculated Subject_Count / 10) * 100 as Pct format=5.1
  from adae
  where TRTEMFL = 'Y'
  group by ARM, AETERM;
quit;

proc report data=ae_summary headline headskip;
  column ARM AETERM Subject_Count Pct;
  define ARM / group "Treatment Group";
  define AETERM / display "Reported Adverse Event";
  define Subject_Count / display "Patient Count (n)";
```